Equilibrium Is Not a Floor: Safe Exploitation and Adaptive Play in Three-Player Poker

Draft v15. Adds the identifier's memory tradeoff (§6.5, Figure 5) and corrects §6.4's recommendation: selection with forgetting has not yet been tested against steady opponents.

Draft v14. Restructured around four questions: why adapt (§4), how much is safe (§5), how to adapt in play (§6), and what happens when opponents adapt back (§7). New since v13: the full grid of independent biases for all nine type pairs and three seats under the pessimistic punisher; the finding that a model's own restricted responses are not the safe frontier, with a re-solve test; bank selection; forgetting and fixed-share memories across ten switching scenarios; a dose-response for opponents who learn; and the three-way chase with restricted-response adapters. Cut: the fixed-alpha identifier study, linear mixing (one sentence remains), and several v13 subsections.

Tags: **[EXACT]** exact game-tree evaluation on the 5,000-iteration baseline. **[SIM]** simulated sessions with exact expected payoffs. All payoffs are chips per hand, relative to the seat's own equilibrium value unless stated otherwise.

Abstract

In two-player zero-sum games, safe opponent exploitation is well understood: restricted Nash responses trace the best possible tradeoff between profiting from an opponent's mistakes and becoming exploitable, measured against the game's value. Neither tool carries over to three players, where there is no game value and an equilibrium strategy guarantees nothing. We study the problem exactly in three-player Leduc Hold'em, against opponents built by biasing an equilibrium toward raising, calling or folding, each with its own degree of bias. Equilibrium play is not a floor: two self-interested opponents who keep learning converge to an equilibrium that costs a fixed equilibrium player 0.39 chips/hand. A three-player restricted Nash response nevertheless achieves **safe exploitation** — at least 0.05 chips/hand more profit with no material loss of worst-case safety — in **288 of 297** settings (median extra profit 0.60), failing mainly against lone deviators. But a model's own restricted responses are often not the best safe strategy: in 41% of settings, a strategy solved for a *different* model earns at least 0.05 more, and the largest gap survives re-solving. Selecting from a bank of such strategies, safely against every model the posterior finds plausible, earns 1.04–1.9 times as much as playing the model's own response against steady opponents; against opponents who switch style, how the posterior forgets matters more than the policy, and no forgetting rate is

both accurate on steady opponents and fast after a switch. Finally, a little trust protects: against two opponents who keep learning, every restricted response with trust $p \leqslant 0.5$ loses less than equilibrium play, and every one with $p \geqslant 0.6$ loses more.

1. Introduction

An equilibrium strategy cannot be exploited but leaves money on the table against weak opponents; a best response wins the most against a specific opponent but can be exploited in turn. In two-player zero-sum games this tradeoff has a well-developed theory. Restricted Nash responses (Johanson, Zinkevich & Bowling, 2007) trace its Pareto frontier, ε-safe strategies bound worst-case loss (McCracken & Bowling, 2004), and safe exploitation risks only what has already been won relative to the game value (Ganzfried & Sandholm, 2015).

Every one of those guarantees rests on the minimax theorem. With three players there is no game value to be safe relative to, an equilibrium strategy guarantees nothing, and one opponent's mistake need not become your gain. Measuring the tradeoff requires a game small enough for exact best responses and large enough that three players genuinely interact. Three-player Leduc Hold'em is that game. We ask four questions:

1. **Why adapt?** Because equilibrium play is not a floor (§4).
2. **How much can be taken safely?** Almost always some, mapped over a grid of independent biases for every type pair and seat — and, surprisingly, a model's own restricted responses are often not the best way to take it (§5).
3. **How should a player adapt in play?** By selecting from a bank of strategies, safely against every model the posterior finds plausible, with a memory suited to how opponents change (§6).
4. **What if opponents adapt back?** A little trust protects; a lot is ruinous (§7).

2. Related work

Robust and safe exploitation. Restricted Nash responses pin the opponent to a model with probability p and otherwise let it best-respond, tracing a Pareto-optimal frontier between gain and exploitability (Johanson et al., 2007); data-biased responses weight the pinning by data confidence (Johanson & Bowling, 2009). ε-safe strategies (McCracken & Bowling, 2004) and safe opponent exploitation (Ganzfried & Sandholm, 2015) bound risk relative to the game value; Jeary & Turrini (2023) recast the value as the worst case of one's own strategy so that guarantees survive approximate equilibria. All are two-player zero-sum. Ganzfried's safe

equilibrium (2023) generalises restricted responses to multiplayer games and is the closest theoretical relative of our construction.

Choosing among precomputed responses. Bard, Johanson, Burch & Bowling (2013) build a portfolio of robust responses offline and choose among them online in heads-up limit hold'em, estimating each one's payoff from play rather than from an explicit model. Bayesian policy reuse (Rosman, Hawasly & Ramamoorthy, 2016) and its extensions to opponents who switch strategy — BPR+ (Hernandez-Leal et al., 2016), switch detection (Hernandez-Leal et al., 2017) and Deep BPR+ (Zheng et al., 2018) — choose a policy from a library using a belief over opponent types. Our bank selection (§6.3) combines the two lines with a safety constraint: it uses an explicit posterior, and requires the chosen strategy to be safe against every type the posterior considers plausible, in a three-player game.

Opponent modelling. Leduc Hold'em was introduced with Bayesian opponent modelling (Southey et al., 2005); posterior-based exploitation in imperfect-information games is studied by Ganzfried & Sun (2018). Our identifier is Bayesian reasoning over parameterised hypothesised types (Albrecht, Crandall & Ramamoorthy, 2016; Albrecht & Stone, 2017; survey in Albrecht & Stone, 2018). The only multiplayer precedent we found is Ganzfried, Wang & Chiswick (2024), in three-player Kuhn poker.

Multiplayer equilibrium. CFR (Zinkevich et al., 2008) has no Nash guarantee beyond two players, though it has produced strong three-player (Abou Risk & Szafron, 2010) and superhuman six-player agents (Brown & Sandholm, 2019). Three-player Kuhn poker already has a parameterised family of equilibria (Szafron, Gibson & Sturtevant, 2013), which foreshadows the equilibrium selection effects we measure. That one opponent's mistake can benefit a third player is poker folklore (Morton's theorem); we quantify it.

3. Setup

Game. Three-player Leduc Hold'em in OpenSpiel (Lanctot et al., 2019): 8 cards across 4 ranks, 25,800 information sets, 1,831,601 game-tree nodes — small enough for exact evaluation.

Equilibrium baseline. CFR+ (Tammelin, 2014) for 5,000 iterations; nash_conv 0.0023. Equilibrium values by seat: −0.095, −0.009, +0.104 (seat 0 acts first in every betting round).

Opponents. Each opponent tilts the equilibrium toward one action wherever it is legal: $\sigma'(a \mid I) = (1 - \alpha)\,\sigma^*(a \mid I) + \alpha\,[a = \text{favoured}]$. Types: **O** equilibrium, **R** over-raiser, **C** over-caller, **F** over-folder. Each opponent has its own α; the main grid uses $\alpha \in \{0, 0.2, 0.4, 0.6\}$ for each, giving 99 settings per seat for all nine type pairs (an opponent with $\alpha = 0$ *is* the equilibrium player). α is not comparable across types: per unit of α, an over-raiser deviates on 1.41 decisions per hand, an over-caller on 0.89, an over-folder on 0.28.

Metrics.

- **Gain:** our expected payoff against the biased pair, relative to equilibrium value.
- **Worst case:** our expected payoff if *either* opponent abandons its model and plays an exact best response to our strategy, maximising its own payoff, while the other keeps its model; the worse of the two. The punisher is **pessimistic**: among responses within 0.01 chips of its best at each decision, it picks the one worst for us (§4.3 shows why tie-breaking matters).
- **Baseline:** the better of two equilibrium-level strategies on each axis — the CFR table and the $p = 0$ restricted response (§3, Methods).
- **Safe exploitation:** some trust $p > 0$ earns at least **0.05** more than the baseline, with a worst case at most **0.05** below the baseline's. §5.5 treats the 0.05 buffer as a dial.

Methods.

- **Three-player restricted Nash response (RNR).** In a modified game, each opponent independently follows its model with probability p and is otherwise *free* to maximise its own payoff; we observe neither mode. Solved with CFR+ for 600 iterations (best-response gaps in the modified game below 0.008). At $p = 0$ the modified game is the real game, so the $p = 0$ response is an independent, shorter equilibrium solve. Each setting is solved at $p = 0$, 0.25 and 0.5, then refined by an adaptive search toward the largest safe p (up to 0.9).
- **Engine.** A vectorised tree engine for CFR on modified games, best responses and exact evaluation, verified against OpenSpiel (§8).
- **Identifier.** An exact Bayesian posterior over each opponent's type and α (the equilibrium player plus R, C, F at $\alpha = 0.05, \ldots, 0.6$), marginalising over unseen cards. Optionally it **forgets** (old evidence discounted by 0.98 per hand, a memory of about 50 hands) or uses **fixed-share** (a 2% chance per hand that an opponent has switched).

4. Why adapt: equilibrium is not a floor [EXACT]

In two-player zero-sum games, an opponent's deviation can only help an equilibrium player. In three players, none of this holds:

- **Deviating opponents can beat a fixed equilibrium player.** Playing CFR against an over-raiser and an over-caller at $\alpha = 1.0$ loses 0.69 chips/hand.
- **Mistakes go to the third player.** With an over-raiser ($\alpha = 0.35$) in seat 1 and an equilibrium player in seat 2, the raiser loses 0.33; we gain 0.13 and the other opponent 0.20. Adapting is how one claims a larger share.
- **An almost-indifferent opponent is a kingmaker.** With everyone at equilibrium, an opponent restricted to responses within 0.001 chips of its best can still move our payoff between −0.23 and +0.11 — deciding, at no cost to itself, which of the other two pays.

- **Learning opponents reach a second equilibrium.** Holding our CFR strategy fixed, two self-interested opponents learning simultaneously, without collusion, converge from three different starts to the same profile (gaps below 0.0012): **we lose 0.39**, seat 1 gains 0.14 and seat 2 gains 0.25. An independently computed equilibrium-level strategy (p = 0) fares almost as badly (−0.36).
- **Over-raisers make equilibrium play punishable.** Over the grid, the baseline's own worst case falls steadily with a raiser's α (to −0.30 at α = 0.6), while next to over-folders it stays safe.

5. How much is safe [EXACT, pessimistic punisher]

5.1 Headline

setting	safe exploitation	median extra profit	median largest safe p
both opponents biased equally, α = 0.2	25 of 27	+0.25	0.70
α = 0.4	**27 of 27**	+0.72	0.70
α = 0.6	**27 of 27**	+1.51	0.70
full grid, seat 0 / 1 / 2	95 / 98 / 95 of 99	+0.51 / +0.58 / +0.66	0.6 / 0.7 / 0.6
full grid, all seats	**288 of 297**	**+0.60**	0.60

Equal-bias rows count the nine type pairs at three seats. Switching from the exact to the pessimistic punisher changes only 3 of 297 verdicts (all to "safe"). Re-solving 24 of the best safe points with twice the iterations or a new random start left 22 verdicts unchanged (median changes: profit 0.001, worst case 0.006).

Figure 1 (`fig_heatmap_extra_s0.pdf`): safe extra profit over both opponents' biases, seat 0, one panel per type pair. Profit grows with both biases; tables with an over-folder stay pale. Companion maps of the largest safe p and for seats 1–2 are in the appendix.

Who can be trusted. Over-callers can be trusted almost fully: every over-caller pair (CC) allows p = 0.9 at seat 0, and tables with an over-caller mostly allow 0.8–0.9. An over-folder in seat 1 caps trust at 0.2–0.4 in most settings.

5.2 The safe region ends abruptly

Figure 2 (`fig_frontier_dense.pdf`): profit against worst case as p rises from 0 to 0.9, for three settings solved at nine or more values of p. With two over-folders (0.4, 0.4), trust initially improves profit *and* safety — the worst case rises to 0.19 above the baseline's at p = 0.25 — and the region stays safe to p = 0.5 (+0.49). The next step, p = 0.6, more than doubles profit (+1.07) but drops the worst case to 0.31 below the baseline's: a switch, not a slope. A lone over-folder offers almost nothing safe (+0.08 at p = 0.3). A lone over-caller stays safe all the way to p = 0.9 (+0.29).

5.3 Where safe exploitation fails

The nine failures are not scattered at random.

- **Six are thin tables, not unsafe ones.** Safe trust exists, but the best safe profit is +0.01 to +0.046, just under the 0.05 materiality threshold. All six have mild biases or a lone deviator.
- **Lone deviators are hardest, and position matters.** With one deviator beside an equilibrium player, safe exploitation holds in 18 of 18 settings for an over-raiser (median +0.45), 16 of 18 for an over-caller (+0.32) and 14 of 18 for an over-folder (+0.11). Every lone-deviator failure occurs when the deviator acts *right before us* (over-callers 7 of 9 succeed in that position, over-folders 5 of 9; all 9 succeed in every type when the deviator acts right after us). The player right before us acts last after we bet, seeing both our action and the other player's response — the strongest position from which to call down the bluffs that exploit a folder. Accordingly, safety against a lone over-folder breaks at p ≈ 0.06–0.31 when it acts before us, against 0.25–0.7 when it acts after.
- **Against over-callers, the bystander punishes.** At the first unsafe trust against a lone over-caller acting before us, the harder punisher is usually the *equilibrium* player: exploiting a caller means betting more for value and bluffing less, and that shift is what the bystander exploits.

The position account is a hypothesis: the counts are small (nine settings per position) and the mechanism is untested. Two of the "unsafe" failures miss the tolerance by under 0.04, within re-solve noise.

5.4 A model's own restricted responses are not the safe frontier

In two-player zero-sum games, restricted responses are provably Pareto-optimal: no strategy beats them on both profit and safety. **In three players that guarantee fails, and the data show it failing.** We evaluated every solved strategy in the bank (≈ 456 per setting, seat 0) against every setting's true opponents:

- The true model's own restricted responses are the best safe strategy (within 0.01) in **45%** of the 99 settings. Another strategy earns **at least 0.05 more in 41%**, and at least 0.20 more in 19%. Median best safe profit: +0.51 on the model's own curve, +0.60 anywhere in the bank.
- The better strategies were mostly solved for the same types at different biases (76%).

- The gaps concentrate where the model's own safe region is narrowest — extreme over-folders and lone deviators. Against a lone over-folder at 0.6, the model's own curve earns nothing safely; a strategy solved for a *milder* folder (0.2) at p = 0.7 earns +0.42.

  Figure 3 (`fig_bank_vs_own_frontier.png`): every bank strategy against three sets of true opponents, with the true model's own curve highlighted.

  Not a convergence artifact. Against an over-folder (0.4) and an over-raiser (0.2), the best safe bank strategy (solved for F0.2-R0.4 at p = 0.7) earns +0.36 with margin +0.04, while the true model's own responses earn at most +0.14 while safe. We re-solved both with twice the iterations and from two new random starts:

strategy	profit (4 versions)	safety margin (4 versions)
true model, p = 0.375	+0.133 to +0.136	−0.108 to −0.043
true model, p = 0.5	+0.228 to +0.239	−0.261 to −0.239
other model's response, p = 0.7	**+0.360 to +0.366**	**+0.037 to +0.053**

  Re-solving moved profits by at most 0.02 and margins by at most 0.065 — mostly with the random start, not the iteration count. In a second case (a lone over-raiser) another model's response again earns twice as much (+0.60 against +0.30) but sits exactly at the safety tolerance (−0.050 to −0.053), so we count it as borderline.

  Why. A restricted response is an equilibrium of a modified game, not a solution of "maximise profit subject to a worst-case limit." In two-player zero-sum games the two coincide, because the free opponent behaves exactly like the worst-case punisher. In three players they come apart: the free opponents are self-interested rather than adversarial, our safety measure moves one punisher at a time while the other keeps its style, and the modified game can have several equilibria (which is why random starts move margins more than iterations do).

5.5 Robustness to model error, and safety as a dial

Model error. Each setting's largest-safe-p strategy, scored against every other setting's true opponents (seat 0, pessimistic p*):

true opponents vs model pairs	still safe	still profitable
exact	98	100%	97%
α off by 0.2	239	73%	98%
α off by 0.4	263	63%	99%
α off by 0.6 or more	180	57%	100%
one type wrong	3,963	41%	83%
both types wrong	4,959	29%	25%

An α error leaves the profit intact but erodes safety steadily; a wrong type breaks both. The largest safe p sits at the edge of the safe region, so small model errors push it over.

Safety as a dial. Figure 4 (`fig_risk_budget.pdf`): median best safe profit as the allowed drop in worst case grows. With no added risk at all, the median setting earns +0.49; the 0.05 buffer raises this to +0.57, two-thirds of the +0.87 available with no limit on risk. The spread across settings (middle half +0.25 to +0.83 at zero risk) is far larger than the effect of the budget: which opponents one faces matters more than how much risk one accepts. Equivalently, if each opponent punishes with probability q and we maximise expected profit, the 0.05 rule corresponds to q ≈ 0.4–0.5; believing punishment rare (q ⩽ 0.2) would justify trust near 0.9 at worst-case costs around 0.3–0.4 chips/hand.

6. How to adapt in play [SIM: 20 sessions × 300 hands per policy and scenario]

6.1 Protocol

We sit in seat 0 against fixed opponents and update the posterior every hand and the strategy every 25 hands. Every strategy played comes from the bank. Payoffs are exact expected values against the true opponents. **Exposure** is the share of hands whose strategy's pessimistic worst case is more than 0.05 below the baseline's. The **hindsight benchmark** is the bank strategy earning most against the true opponents while staying safe against them.

6.2 Policies

- **step τ:** p = 0 until confidence exceeds τ, then the most likely setting's largest safe p.

- **backed-off variants:** one solved point below that p (*notch*), or the largest p safe against all settings within 0.2 of the model (*neighbourhood*).
- **selection (sel):** take the smallest set of settings holding 90% of the posterior; among *all* bank strategies, keep those safe against every one of them; play the one with the highest posterior-weighted profit (p = 0 if none qualifies). Uncertainty makes it cautious automatically; no confidence threshold is needed.
- **best response:** an unrestricted best response to the most likely model once confident, as a reference.

6.3 Steady opponents: selection wins

opponents	step 0.5	selection	hindsight best	best response
over-folder 0.4 + over-raiser 0.2	+0.28	**+0.55**	+0.59	+1.04 (92% exposed)
lone over-raiser 0.4	+0.40	**+0.62**	+0.76	+0.95 (92%)
over-caller 0.2 + over-raiser 0.6	+1.16	**+1.21**	+1.29	+1.70 (92%)
off-grid: F 0.3 + R 0.3	+0.42 (8% exposed)	**+0.67**	+0.82	+1.01 (92%)

Selection earns 1.04–1.9 times as much as step, reaching 81–94% of the hindsight benchmark, and is never exposed (one margin sits exactly at the tolerance). Committing early helps: step at $\tau = 0.5$ matches or beats $\tau = 0.9$. Backing off from the largest safe p costs 20–60% of step's profit without reducing exposure, because exposure here comes from model *error*, which a smaller p does not fix reliably. The best response earns 1.4–3.7 times step's profit but is exposed 92% of the time, with worst cases of −1.4 to −2.8.

6.4 Opponents who switch: memory matters more than policy

Ten scenarios: eight form a 2 × 2 × 2 design — what changes (type, e.g. F0.4 → R0.4, or degree, e.g. F0.2 → F0.6) × one or two deviators × which opponent switches (the one acting right after or right before us) — all on the grid, switching at hand 150 of 300; two earlier scenarios behave consistently. Means over the eight (never adapting earns +0.20; hindsight best +0.80):

memory	step 0.5	selection	best response
none	+0.52, 17% exposed	+0.58, **26% exposed**	+0.88, 90% exposed
forgetting (≈ 50 hands)	+0.55, 8%	+0.55, **8%**	+0.98, 88%
fixed-share (2%)	+0.54, 6%	+0.45, **3%**	+0.95, 76%

Without memory, selection is more exposed than step (worse in four of eight scenarios): it plays more aggressive strategies, so a stale model has further to fall. Forgetting removes the difference: both policies reach 68–69% of the hindsight benchmark at 8% exposure. Fixed-share keeps the posterior from ever becoming certain, which makes selection the safest policy of all, at the lowest profit. The hardest case for every safe policy is a type switch by the opponent acting right before us (selection 50% exposed without memory, 33% with forgetting, 8% with fixed-share), echoing §5.3.

Recommendation, provisionally. Selection with forgetting: as good as step against switching opponents. Against steady opponents selection is far better *without* forgetting (§6.3); whether it keeps that advantage *with* forgetting is untested, and §6.5 suggests a cost, since forgetting lowers steady classification accuracy. **[PENDING: steady scenarios with forgetting and fixed-share.]**

6.5 How fast to forget

With forgetting factor λ, the identifier remembers about 1/(1 − λ) hands. Steady classification needs enough evidence within that window to tell the true table from its nearest rival, which sets a floor on memory; after a switch, the old evidence decays like λ^t, so the time to catch the switch grows with memory. We measured both, with seat 0 playing p = 0 throughout so that only identification is tested: **steady accuracy** is the share of checkpoints, from hand 100 until any switch, at which the most likely table is the true one with confidence above 0.5 (thirteen scenarios, 20 sessions each); **detection delay** is the number of hands after a switch until the same holds for the new table.

memory (hands)	all	~100	~50	~33	~20	~14	~10
steady accuracy	81%	79%	**73%**	**65%**	47%	33%	20%
switch caught after (median hands)	never*	130	**75**	**55**	35	32	35

* Over half the sessions never catch the switch within the 150 remaining hands.

Figure 5 (`fig_memory_tradeoff.pdf`) plots the tradeoff. Three things stand out. Without forgetting, most switches are never caught: 150 hands at the old table outweigh the hands

that follow. Below a memory of about 20 hands, detection gets no faster (about 33 hands) while steady accuracy collapses; the limit there is the evidence needed to become confident about the new table, not memory. The knee lies at memories of 33–50 hands ($\lambda = 0.97$–0.98), which is why $\lambda = 0.98$ served well in §6.4.

The right memory depends on the opponent. Tables with an over-raiser tolerate short memories (a lone over-raiser keeps 89% accuracy at memory ≈ 33), while folder tables do not (a lone over-folder at 0.2 falls from 87% to 47%), as expected from how rarely folders reveal themselves (§3). Some tables are hard at any memory: an over-folder with an over-caller reaches only 51% even without forgetting.

7. When opponents adapt back [EXACT]

Holding our strategy fixed in seat 0, both opponents learn simultaneously from a uniform start, each maximising its own payoff (300 iterations; best-response gaps below 0.005). Equilibrium play loses 0.39 (§4). Restricted responses:

setting (seat 0)	p ≈ 0.2–0.25	p = 0.5	p = 0.6–0.8
over-raisers 0.4 / 0.4	−0.11	−0.21	−0.53 (0.7), −1.28 (0.8)
over-folder 0.4 / over-raiser 0.2	−0.02	−0.18	
lone over-raiser 0.4	−0.03	−0.22	
over-caller 0.2 / over-raiser 0.6	−0.13	−0.21	−0.52 (0.6)
over-folders 0.4 / 0.4	−0.01	−0.38	−1.68 (0.7)
lone over-folder 0.4	−0.25 (p = 0.2)	−0.16	

Every restricted response with $p \leq 0.5$ loses less than equilibrium play; every one with $p \geq 0.6$ loses more. Within each setting the loss grows with p, except against the lone over-folder, whose frontier is irregular (§5.2). Why modest trust *beats* equilibrium is open; one hypothesis is that equilibrium strategies are nearly indifferent at many decisions, which two learners can exploit together, while a model-tilted strategy breaks those ties.

This ties the paper together: how much to trust depends on whether opponents will adapt. Against steady opponents, the largest safe p is the right target (§6.3); against opponents who learn, it is the wrong one.

When every player adapts. In a three-way chase (ten sessions of 400 hands; each player starts with a style and adapts), every player becomes fully confident in a mild tilt of each opponent, and the two opponents watching the same player share the same belief — yet each player's actual strategy lies 0.1–0.3 (reach-weighted total variation) from the nearest tilt, against 0.04 for a strategy that is half one tilt and half another. Models built on fixed styles stop describing anyone. Adapting safely still keeps the table stable: with restricted-response adapters, every player stays within 0.26 chips/hand of its equilibrium value after hand 50, settling into one of four configurations; with Bayesian best-response adapters, payoffs swing between −0.97 and +1.56 after hand 50 and never settle. Identifiers whose hypotheses include adaptive players are the natural next step.

8. Validity checks

check	result
Engine vs OpenSpiel	identical to 6 decimals for expected values and best responses
Same baseline throughout	saved strategies re-evaluate with 0.0 difference on the 5,000-iteration table
Solution stability	24 best safe points re-solved: 22 verdicts unchanged; median changes 0.001 (profit), 0.006 (worst case)
Own-model vs bank (§5.4)	re-solved with 2× iterations and two random starts; the gap survives in every version
Pessimistic punisher	changes 3 of 297 verdicts, all toward "safe"
Tie-breaking at equilibrium	our payoff can range over ±0.23 (§4); single-value claims avoided
Second equilibrium	three starts converge to the same profile, gaps < 0.0012
Identifier	matches brute force over every deal (3e−15); tilt formula equals the opponent construction (2e−16)
Projection (§7)	an exact tilt projects onto itself at distance 0; between grid points, 0.013

9. Discussion and limitations

What carries over from two players, and what does not. Restricted responses still find safe exploitation almost everywhere, but they are no longer the frontier, equilibrium is no longer a floor, and safety depends on position and on which opponent punishes. Safety must be measured against equilibrium-level baselines rather than a game value, and that measurement is sensitive to tie-breaking and to small strategy differences; we calibrated against both.

Limitations. One small game. Opponents are tilts of one equilibrium; real opponents deviate in richer ways, and the chase shows that adaptive players are not tilts at all. The bank matrix and selection are computed for seat 0 only. Safety is measured against one punisher at a time; §7 shows two learners can do worse. Selection is scored with the same evaluations it selects by, so it cannot see solution noise; the re-solve test (§5.4) bounds that noise at about 0.065. The position account (§5.3) and the mechanism behind §7 are hypotheses. A single forgetting rate cannot serve every opponent (§6.5); a memory that depends on the suspected type, or a long memory reset by a detector when recent hands fit the current model badly (in the spirit of Hernandez-Leal et al., 2017), could do better. The second is the simpler to build, because the posterior is joint over both opponents. Sessions are simulated with exact expected payoffs, not realised chips.

10. Conclusion

In three-player poker, equilibrium play offers no floor, so a player has reason to adapt; safe exploitation is available in 97% of the settings we mapped; but the two-player recipe for it no longer gives the best strategies, and how far to trust a model depends on whether opponents will trust theirs.

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✓ details checked; ⚠ details to confirm. Every reference must have at least 3 citations; those marked **[confirm ≥ 3 citations]** should be checked on Google Scholar before submission.

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  Dropped from v13: Hoehn et al. (2005) and Southey, Hoehn & Holte (2009), no longer cited in the text.

Appendix

Figures. Fig. 1 `fig_heatmap_extra_s0.pdf`; Fig. 2 `fig_frontier_dense.pdf`; Fig. 3 `fig_bank_vs_own_frontier.png`; Fig. 4 `fig_risk_budget.pdf`; Fig. 5 `fig_memory_tradeoff.pdf`. Appendix: `fig_heatmap_pmax_s0.pdf` (largest safe p) and both maps for seats 1–2 (`fig_heatmap_*_s1.pdf`, `fig_heatmap_*_s2.pdf`).

Reproduction (`rnr_package/`). Solves: `rnr_general.py`, `overnight_grid.py`. Scoring: `reeval_het.py` (pessimistic), `stability_het.py`, `own_frontier_check.py` (§5.4 re-solves), `cross_eval.py` (model error, bank matrix, envelope). Play: `q2_grid_identify.py`, `adapt_policies.py` (§6), `memory_sweep.py` (§6.5), `gaps_check.py` (§4, §7), `session_sim.py` (chase). `cfr_path.py` fixes the CFR table every script reads.